

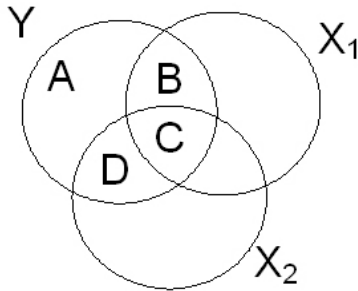
Statistics Homework 5:

Multiple Regression

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Part A: Venn diagram

Consider the Venn diagram below and identify which statistic is being described in each of the following items. Answer each item by typing an abbreviation from the Acceptable Abbreviations Page.



1. What is the name of the statistic that is equal to $\frac{B}{A+B}$?
2. What is the name of the statistic that is equal to $\sqrt{\frac{B+C}{A+B+C+D}}$?
3. What is the name of the statistic that is equal to $\sqrt{\frac{B}{A+B+C+D}}$?
4. What is the name of the statistic that is equal to $\frac{B+C}{A+B+C+D}$?
5. What is the name of the statistic that is equal to $\sqrt{\frac{B}{A+B}}$?
6. What is the name of the statistic that is equal to $\frac{B+C+D}{A+B+C+D}$?
7. What is the name of the statistic that is equal to $\frac{B}{A+B+C+D}$?
8. What is the name of the statistic that pertains to area “B” when none of the variables in the equation are standardized?
9. What is the name of the statistic that is equal to $\sqrt{\frac{B+C+D}{A+B+C+D}}$?
10. What is the name of the statistic that pertains to area “B” when all of the variables are standardized prior to analysis?

Part B: Definitions

Match the term with the definition. (Note, some terms will be used more than once)

11. When a bivariate correlation between two variables is smaller than the partial correlation between those variables.
12. The resulting beta when all the variables are turned into z-scores prior to analysis.
13. A high correlation between two predictor variables in a regression equation.
14. A regression parameter estimate that can be interpreted as having a scale that is **similar** to a correlation (but not identical with a correlation).
15. Three biased statistics.
16. A regression parameter estimate for which a value of .8 could indicate a SMALLER effect than a value of .2
17. A parameter estimate in multiple regression that **is** a correlation.
18. A statistic that indicates how much additional variance is explained when one set of predictors is added to another set.
19. The resulting beta when all the variables are residuals that have been turned into z-scores.

- A Collinearity
 B Partial correlation
 C Delta R^2
 D Suppressor effect

- E R^2 R r^2
 F Unstandardized beta
 G Standardized beta

Part C: ANOVA table

The ANOVA table below is from a multiple regression analysis. Use the data in the table to determine the following:

ANOVA

20. R^2
21. standard error of estimate
22. number of predictors
23. sample size

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12.777	4	3.194	10.138	.000
	Residual	46.632	148	.315		
	Total	59.409	152			

Another ANOVA Table is depicted below. Use the data in the table to determine the following:

24. R^2
25. standard error of estimate
26. number of predictors
27. sample size

ANOVA

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	158.278	6	26.380	15.949	.000
	Residual	181.944	110	1.654		
	Total	340.222	116			

(Note: question 28 does not exist)

Part D: Calculating b_1

Consider the following target regression equation:

[target equation] $y = b_0 + b_1x_1 + b_2x_2 + e$

Choose from the options listed in the key and fill in the boxes to describe the steps for calculating b_1 (the first slope) in the target equation.

To calculate b_1 in the target equation, the first step is to calculate results for the equation:

$$y = b_0 + \boxed{29} \boxed{30} + \boxed{31}$$

The second step is to calculate results for the equation:

$$\boxed{32} = b_0 + \boxed{33} \boxed{34} + \boxed{35}$$

The third step is to calculate results for the equation:

$$\boxed{36} = b_0 + \boxed{37} \boxed{38} + \boxed{39}$$

Key			
A	x_1	G	$e_{y.x2}$
B	x_2	H	$e_{y.x1}$
C	y	I	$e_{x1.x2}$
D	b_1	J	$e_{x1.y}$
E	b_2	K	$e_{x2.x1}$
F	e	L	$e_{x2.y}$

40. In the above equations, which box number did you identify as being equal to b_1 from the target equation?

Part E: Cognition and communication behavior in couples

When couples experience relationship conflicts, two types of cognition (types of thinking) may be especially important. First, attributions have to do with how much a person blames the other partner for a conflict. Second, expectancies have to do with how much a person thinks the other partner is likely to do something negative behavior in the future. Although these two types of cognition may be correlated with each other and also correlated with ratings of relationship satisfaction, they are also expected to be distinct. Thus, each type of cognition is expected to predict conflict communication behavior after controlling for any overlap these variables share with each other and with relationship satisfaction.

Sanford (2006) conducted a study in which a sample of 76 recently-married, heterosexual couples completed a series of assessments in a communication laboratory. For each assessment, couples first identified an area of conflict in their relationship. Then, each partner completed a series of questionnaires that included: (a) an attribution scale assessing the extent to which respondents blamed their partners for the identified conflict, (b) an expectancies scale assessing the extent to which respondents expected their partners to use negative communication in future conversations, and (c) a relationship satisfaction scale called the Quality Marriage Index. Next, the couple was taken to a room and asked to discuss the conflict, and their conversations were video recorded. The videos were later coded by a research team for the extent to which each partner used negative or hostile communication during the discussion. Regression was used to investigate the extent to which negative expectancies, negative attributions, and relationship satisfaction predicted communication behavior. Results for wives are printed on the next page.

Descriptive Statistics

	Mean	Std. Deviation	N
w_neg_com	.3644	.33485	76
w_expect	3.0395	1.54167	76
w_att	3.0811	1.08849	76
w_qmi	34.2039	6.85866	76

Data from Wives

w_neg_com = wife negative communication
w_expect = wife negative expectancies
w_att = wife negative attributions
w_qmi = wife marital satisfaction score on the quality marriage index.

Correlations

		w_neg_com	w_expect	w_att	w_qmi
w_neg_com	Pearson Correlation	1.000	.533	.564	-.370
	Sig. (1-tailed)		.000	.000	.001
	N	76	76	76	76
w_expect	Pearson Correlation	.533	1.000	.508	-.424
	Sig. (1-tailed)	.000		.000	.000
	N	76	76	76	76
w_att	Pearson Correlation	.564	.508	1.000	-.365
	Sig. (1-tailed)	.000	.000		.001
	N	76	76	76	76
w_qmi	Pearson Correlation	-.370	-.424	-.365	1.000
	Sig. (1-tailed)	.001	.000	.001	
	N	76	76	76	76

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.639 ^a	.409	.384	.26274

a. Predictors: (Constant), w_qmi, w_att, w_expect

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	3.439	3	1.146	16.606	.000 ^a
	Residual	4.970	72	.069		
	Total	8.410	75			

a. Predictors: (Constant), w_qmi, w_att, w_expect

b. Dependent Variable: w_neg_com

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-.007	.231		-.032	.974
	w_expect	.065	.024			.009
	w_att	.115	.033			.001
	w_qmi	-.005	.005			.297

a. Dependent Variable: w_neg_com

Questions regarding data from wives on page 4

41. What is the typical distance wives' scores deviated from the mean of y ?
42. What is the typical distance wives' observed scores deviated from their predicted values?
43. What is the correlation between wives' observed scores and their predicted values?

Fill in the blanks below with numerical values so that the sentence gives a clear and accurate interpretation of the beta for x_1 .

44. Y is expected to increase by _____ units if
45. x_1 increases by _____ units and
46. x_2 is _____ and
47. x_3 is _____.

Fill in the blanks below with numerical values so that the sentence gives a clear and accurate interpretation of the beta for x_2 .

48. Y is expected to increase by _____ units if
49. x_1 is _____ and
50. x_2 increases by _____ units and
51. x_3 is _____.

Fill in the blanks below with numerical values so that the sentence gives a clear and accurate interpretation of the beta for x_3 .

52. Y is expected to increase by _____ units if
53. x_1 is _____ and
54. x_2 is _____ and
55. x_3 increases by _____ units.

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

For wives, what is the standardized association between X_1 and Y (without controlling for other variables)?

- | | |
|---|--|
| 56. What is the name of the descriptive statistic? | 58. What is the name of the inferential statistic? |
| 57. What is the value of the descriptive statistic? | 59. What is the value of the inferential statistic? |
| | 60. What is the p value (exactly as printed)? |

For wives, what is the standardized association between X_1 and Y after controlling for other variables in the equation?

- | | |
|---|--|
| 61. What is the name of the descriptive statistic? | 63. What is the name of the inferential statistic? |
| 62. What is the value of the descriptive statistic? | 64. What is the value of the inferential statistic? |
| | 65. What is the p value (exactly as printed)? |

For wives, what is the standardized association between X_2 and Y (without controlling for other variables)?

- | | |
|---|--|
| 66. What is the name of the descriptive statistic? | 68. What is the name of the inferential statistic? |
| 67. What is the value of the descriptive statistic? | 69. What is the value of the inferential statistic? |
| | 70. What is the p value (exactly as printed)? |

For wives, what is the standardized association between X_2 and Y after controlling for other variables in the equation?

- | | |
|---|--|
| 71. What is the name of the descriptive statistic? | 73. What is the name of the inferential statistic? |
| 72. What is the value of the descriptive statistic? | 74. What is the value of the inferential statistic? |
| | 75. What is the p value (exactly as printed)? |

For wives, what is the standardized association between X_3 and Y (without controlling for other variables)?

- | | |
|---|--|
| 76. What is the name of the descriptive statistic? | 78. What is the name of the inferential statistic? |
| 77. What is the value of the descriptive statistic? | 79. What is the value of the inferential statistic? |
| | 80. What is the p value (exactly as printed)? |

For wives, what is the standardized association between X_3 and Y after controlling for other variables in the equation?

- | | |
|---|--|
| 81. What is the name of the descriptive statistic? | 83. What is the name of the inferential statistic? |
| 82. What is the value of the descriptive statistic? | 84. What is the value of the inferential statistic? |
| | 85. What is the p value (exactly as printed)? |

For wives, what is the total percent of variance in Y explained by all the predictors taken together?

- | | |
|---|--|
| 86. What is the name of the descriptive statistic? | 88. What is the name of the inferential statistic? |
| 87. What is the value of the descriptive statistic? | 89. What is the value of the inferential statistic? |
| | 90. What is the p value (exactly as printed)? |

You just completed the section of question regarding data from wives.

Results for husbands are on the next page.

Continued. . .

Descriptive Statistics

	Mean	Std. Deviation	N
h_neg_com	.2442	.23536	76
h_expect	3.0493	1.26361	76
h_att	2.9211	.97733	76
h_qmi	34.1447	6.17242	76

Data from Husbands

h_neg_com = husband negative communication

h_expect = husband negative expectancies

h_att = husband negative attributions

h_qmi = husband marital satisfaction score on the quality marriage index.

Correlations

		h_neg_com	h_expect	h_att	h_qmi
h_neg_com	Pearson Correlation	1.000	.438	.421	-.509
	Sig. (1-tailed)	.	.000	.000	.000
	N	76	76	76	76
h_expect	Pearson Correlation	.438	1.000	.243	-.414
	Sig. (1-tailed)	.000	.	.017	.000
	N	76	76	76	76
h_att	Pearson Correlation	.421	.243	1.000	-.315
	Sig. (1-tailed)	.000	.017	.	.003
	N	76	76	76	76
h_qmi	Pearson Correlation	-.509	-.414	-.315	1.000
	Sig. (1-tailed)	.000	.000	.003	.
	N	76	76	76	76

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.618 ^a	.381	.356	.18892

a. Predictors: (Constant), h_qmi, h_att, h_expect

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.585	3	.528	14.802	.000 ^a
	Residual	2.570	72	.036		
	Total	4.155	75			

a. Predictors: (Constant), h_qmi, h_att, h_expect

b. Dependent Variable: h_neg_com

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.353	.192		1.844	.069
	h_expect	.044	.019			.023
	h_att	.063	.024			.010
	h_qmi	-.013	.004			.003

a. Dependent Variable: h_neg_com

Questions regarding data from husbands on page 7

91. What is the typical distance people's scores deviated from the mean of y?
92. What is the typical distance people's observed scores deviated from their predicted values?
93. What is the correlation between people's observed scores and their predicted values?
94. After controlling for the effects of other predictors, how many units is y expected to increase when x_1 increases by 1 unit?
95. After controlling for the effects of other predictors, how many standard deviations is y expected to increase when x_1 increases by 1 standard deviation?
96. After controlling for the effects of other predictors, how many units is y expected to increase when x_2 increases by 1 unit?
97. After controlling for the effects of other predictors, how many standard deviations is y expected to increase when x_2 increases by 1 standard deviation?
98. After controlling for the effects of other predictors, how many units is y expected to increase when x_3 increases by 1 unit?
99. After controlling for the effects of other predictors, how many standard deviations is y expected to increase when x_3 increases by 1 standard deviation?

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

For husbands, what is the standardized association between X_1 and Y (without controlling for other variables)?

- | | |
|--|---|
| 100. What is the name of the descriptive statistic? | 102. What is the name of the inferential statistic? |
| 101. What is the value of the descriptive statistic? | 103. What is the value of the inferential statistic? |
| | 104. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between X_1 and Y after controlling for other variables in the equation?

- | | |
|--|---|
| 105. What is the name of the descriptive statistic? | 107. What is the name of the inferential statistic? |
| 106. What is the value of the descriptive statistic? | 108. What is the value of the inferential statistic? |
| | 109. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between X_2 and Y (without controlling for other variables)?

- | | |
|--|---|
| 110. What is the name of the descriptive statistic? | 112. What is the name of the inferential statistic? |
| 111. What is the value of the descriptive statistic? | 113. What is the value of the inferential statistic? |
| | 114. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between X_2 and Y after controlling for other variables in the equation?

- | | |
|--|---|
| 115. What is the name of the descriptive statistic? | 117. What is the name of the inferential statistic? |
| 116. What is the value of the descriptive statistic? | 118. What is the value of the inferential statistic? |
| | 119. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between X_3 and Y (without controlling for other variables)?

- | | |
|--|---|
| 120. What is the name of the descriptive statistic? | 122. What is the name of the inferential statistic? |
| 121. What is the value of the descriptive statistic? | 123. What is the value of the inferential statistic? |
| | 124. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between X_3 and Y after controlling for other variables in the equation?

- | | |
|--|---|
| 125. What is the name of the descriptive statistic? | 127. What is the name of the inferential statistic? |
| 126. What is the value of the descriptive statistic? | 128. What is the value of the inferential statistic? |
| | 129. What is the p value (exactly as printed)? |

For husbands, what is the total percent of variance in Y explained by all the predictors taken together?

- | | |
|--|---|
| 130. What is the name of the descriptive statistic? | 132. What is the name of the inferential statistic? |
| 131. What is the value of the descriptive statistic? | 133. What is the value of the inferential statistic? |
| | 134. What is the p value (exactly as printed)? |

Part F: Emotion and communication behavior in couples

In addition to cognition, it may be important to understand the emotions that couples experience during relationship conflicts. Two potentially important types of emotion include hard emotion and soft emotion. Hard emotion includes feelings of anger, contempt, and aggravation, and this type of emotion is associated with seeking power and control and it may motivate people to protect themselves against partners that are perceived as harmful or neglectful. In contrast, soft emotion includes feeling sad, hurt, or disappointed, and this type of emotion is unique in that it involves expressions of vulnerability. On the one hand both of these emotions are similar in that they are negative emotions that people often experience during relationship conflicts. In other words, they both share a common component that could be labeled “non-specific negative emotion.” On the other hand, each type of emotion is expected to contain a unique component and the unique components are expected to have particular effects on conflict communication. Because the unique component of hard emotion involves power and control, it is expected to predict conflict escalation. Because the unique component of soft emotion involves expressions of vulnerability, it is expected to facilitate conflict resolution.

Sanford (2007) conducted a study in which 81 couples were video recorded as they discussed areas of relationship conflict. Each video was coded by two different teams of research assistants. The first team rated the extent of non-specific negative emotion and non-specific positive emotion in each partner. The second team rated the extent of hard emotion and the extent of soft emotion in each partner. This team also rated the extent of negative or hostile communication used by each partner. Regression was used to identify the extent to which hard and soft emotion explained variance in communication after controlling for variance that could be explained by non-specific positive and negative emotion. Results for wives are printed on the next two pages.

Descriptive Statistics

	Mean	Std. Deviation	N
w_pos_com	.7796	.37813	81
w_neg_emo	.2969	.30333	81
w_pos_emo	.5727	.40550	81
w_hard	.0143	.73866	81
w_soft	.1366	.61712	81

Data from Wives

w_pos_com = wife positive communication

w_neg_emo = wife non-specific negative emotion

w_pos_emo = wife non-specific positive emotion

w_hard = wife hard emotion

w_soft = wife soft emotion

Correlations

		w_pos_com	w_neg_emo	w_pos_emo	w_hard	w_soft
w_pos_com	Pearson Correlation	1.000	-.584	.474	-.648	-.154
	Sig. (1-tailed)	.	.000	.000	.000	.086
	N	81	81	81	81	81
w_neg_emo	Pearson Correlation	-.584	1.000	-.463	.794	.693
	Sig. (1-tailed)	.000	.	.000	.000	.000
	N	81	81	81	81	81
w_pos_emo	Pearson Correlation	.474	-.463	1.000	-.402	-.409
	Sig. (1-tailed)	.000	.000	.	.000	.000
	N	81	81	81	81	81
w_hard	Pearson Correlation	-.648	.794	-.402	1.000	.507
	Sig. (1-tailed)	.000	.000	.000	.	.000
	N	81	81	81	81	81
w_soft	Pearson Correlation	-.154	.693	-.409	.507	1.000
	Sig. (1-tailed)	.086	.000	.000	.000	.
	N	81	81	81	81	81

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.628 ^a	.394	.379	.29801	.394	25.400	2	78	.000
2	.777 ^b	.604	.583	.24424	.209				.000

a. Predictors: (Constant), w_pos_emo, w_neg_emo

b. Predictors: (Constant), w_pos_emo, w_neg_emo, w_soft, w_hard

ANOVA^c

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4.511	2	2.256	25.400	.000 ^a
	Residual	6.927	78	.089		
	Total	11.439	80			
2	Regression	6.905	4	1.726	28.939	.000 ^b
	Residual	4.534	76	.060		
	Total	11.439	80			

a. Predictors: (Constant), w_pos_emo, w_neg_emo

b. Predictors: (Constant), w_pos_emo, w_neg_emo, w_soft, w_hard

c. Dependent Variable: w_pos_com

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.813	.084		9.666	.000
	w_neg_emo	-.579	.124	-.464	-4.673	.000
	w_pos_emo	.242	.093	.259	2.609	.011
2	(Constant)	.755	.076		9.867	.000
	w_neg_emo	-.581	.180	-.466	-3.230	.002
	w_pos_emo	.277	.077	.297	3.604	.001
	w_hard	-.211	.061	-.412	-3.439	.001
	w_soft	.306	.062	.500	4.912	.000

a. Dependent Variable: w_pos_com

Questions regarding data from wives on pages 10 and 11

135. Calculate the sum of squares for the explained variance for ΔF .
136. Calculate the degrees of freedom for the explained variance for ΔF .
137. Calculate the mean square for the explained variance for ΔF .
138. What is the mean square error variance for ΔF ?
139. Calculate the ΔF .
140. Calculate the partial correlation for hard emotion (in model 2).
141. Calculate the partial correlation for soft emotion (in model 2).
142. Which variable showed a pattern of results that indicates a suppressor effect?
 - A. positive communication
 - B. non-specific negative emotion
 - C. non-specific positive emotion
 - D. hard emotion
 - E. soft emotion
 - F. none of the above

To answer each of the following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

Note, to save on time and space, the following section begins with questions regarding X_3 , skipping over X_1 and X_2 .

For wives, what is the standardized association between hard emotion and positive communication (without controlling for other variables)?

- | | |
|--|---|
| 143. What is the name of the descriptive statistic? | 145. What is the name of the inferential statistic? |
| 144. What is the value of the descriptive statistic? | 146. What is the value of the inferential statistic? |
| | 147. What is the p value (exactly as printed)? |

For wives, what is the standardized association between hard emotion and positive communication after controlling for all the other variables in the equation?

- | | |
|--|---|
| 148. What is the name of the descriptive statistic? | 150. What is the name of the inferential statistic? |
| 149. What is the value of the descriptive statistic? | 151. What is the value of the inferential statistic? |
| | 152. What is the p value (exactly as printed)? |

For wives, what is the standardized association between soft emotion and positive communication (without controlling for other variables)?

- | | |
|--|---|
| 153. What is the name of the descriptive statistic? | 155. What is the name of the inferential statistic? |
| 154. What is the value of the descriptive statistic? | 156. What is the value of the inferential statistic? |
| | 157. What is the p value (exactly as printed)? |

For wives, what is the standardized association between soft emotion and positive communication after controlling for other variables in the equation?

- | | |
|--|---|
| 158. What is the name of the descriptive statistic? | 160. What is the name of the inferential statistic? |
| 159. What is the value of the descriptive statistic? | 161. What is the value of the inferential statistic? |
| | 162. What is the p value (exactly as printed)? |

For wives, what percent of the variance in Y was explained when just the first set of variables were entered on step one?

- | | |
|--|---|
| 163. What is the name of the descriptive statistic? | 165. What is the name of the inferential statistic? |
| 164. What is the value of the descriptive statistic? | 166. What is the value of the inferential statistic? |
| | 167. What is the p value (exactly as printed)? |

For wives, how much additional variance in Y was explained when the second set of variables were added to the equation on step two?

- | | |
|--|---|
| 168. What is the name of the descriptive statistic? | 170. What is the name of the inferential statistic? |
| 169. What is the value of the descriptive statistic? | 171. What is the value of the inferential statistic? |
| | 172. What is the p value (exactly as printed)? |

For wives, what is the total percent of variance in Y explained by all the predictors taken together?

- | | |
|--|---|
| 173. What is the name of the descriptive statistic? | 175. What is the name of the inferential statistic? |
| 174. What is the value of the descriptive statistic? | 176. What is the value of the inferential statistic? |
| | 177. What is the p value (exactly as printed)? |

You just completed the section of question regarding data from wives. Results for husbands are printed below. Use these results to answer the questions that follow.

Descriptive Statistics

	Mean	Std. Deviation	N
h_pos_com	.8351	.38700	81
h_neg_emo	.2120	.23852	81
h_pos_emo	.4747	.35010	81
h_hard	-.3007	.52043	81
h_soft	-.1751	.45374	81

Data from Husbands

h_pos_com = husband positive communication

h_neg_emo = husband non-specific negative emotion

h_pos_emo = husband non-specific positive emotion

h_hard = husband hard emotion

h_soft = husband soft emotion

Correlations

		h_pos_com	h_neg_emo	h_pos_emo	h_hard	h_soft
h_pos_com	Pearson Correlation	1.000	-.390	.286	-.490	.198
	Sig. (1-tailed)	.	.000	.005	.000	.038
	N	81	81	81	81	81
h_neg_emo	Pearson Correlation	-.390	1.000	-.435	.828	.659
	Sig. (1-tailed)	.000	.	.000	.000	.000
	N	81	81	81	81	81
h_pos_emo	Pearson Correlation	.286	-.435	1.000	-.443	-.368
	Sig. (1-tailed)	.005	.000	.	.000	.000
	N	81	81	81	81	81
h_hard	Pearson Correlation	-.490	.828	-.443	1.000	.480
	Sig. (1-tailed)	.000	.000	.000	.	.000
	N	81	81	81	81	81
h_soft	Pearson Correlation	.198	.659	-.368	.480	1.000
	Sig. (1-tailed)	.038	.000	.000	.000	.
	N	81	81	81	81	81

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.410 ^a	.168	.147	.35741	.168	7.899	2	78	.001
2	.767 ^b	.588	.566	.25485	.420				.000

a. Predictors: (Constant), h_pos_emo, h_neg_emo

b. Predictors: (Constant), h_pos_emo, h_neg_emo, h_soft, h_hard

ANOVA^c

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2.018	2	1.009	7.899	.001 ^a
	Residual	9.964	78	.128		
	Total	11.982	80			
2	Regression	7.045	4	1.761	27.118	.000 ^b
	Residual	4.936	76	.065		
	Total	11.982	80			

a. Predictors: (Constant), h_pos_emo, h_neg_emo

b. Predictors: (Constant), h_pos_emo, h_neg_emo, h_soft, h_hard

c. Dependent Variable: h_pos_com

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.873	.094		9.291	.000
	h_neg_emo	-.531	.186	-.327	-2.854	.006
	h_pos_emo	.158	.127	.143	1.249	.216
2	(Constant)	.977	.098		9.952	.000
	h_neg_emo	-.917	.252	-.565	-3.643	.000
	h_pos_emo	.210	.093	.190	2.269	.026
	h_hard	-.237	.101	-.319	-2.359	.021
	h_soft	.677	.085	.793	7.920	.000

a. Dependent Variable: h_pos_com

Questions regarding data from husbands on pages 13 and 14

178. Calculate the sum of squares for the explained variance for ΔF .
179. Calculate the degrees of freedom for the explained variance for ΔF .
180. Calculate the mean square for the explained variance for ΔF .
181. What is the mean square error variance for ΔF ?
182. Calculate the ΔF .
183. Calculate the partial correlation for hard emotion (in model 2).
184. Calculate the partial correlation for soft emotion (in model 2).

To answer each of following questions, provide 5 pieces of information: (1) the name of the descriptive statistic, (2) the value of the descriptive statistic, (3) the name of the inferential statistic, (4) the value of the inferential statistic, and (5) the p value. When reporting the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page. **Report all p values exactly as printed in the results with NO extra zeros and NO rounding.**

Note, to save on time and space, the following section begins with questions regarding X_3 , skipping over X_1 and X_2 .

For husbands, what is the standardized association between hard emotion and positive communication (without controlling for other variables)?

- | | |
|--|---|
| 185. What is the name of the descriptive statistic? | 187. What is the name of the inferential statistic? |
| 186. What is the value of the descriptive statistic? | 188. What is the value of the inferential statistic? |
| | 189. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between hard emotion and positive communication after controlling for all the other variables in the equation?

- | | |
|--|---|
| 190. What is the name of the descriptive statistic? | 192. What is the name of the inferential statistic? |
| 191. What is the value of the descriptive statistic? | 193. What is the value of the inferential statistic? |
| | 194. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between soft emotion and positive communication (without controlling for other variables)?

- | | |
|--|---|
| 195. What is the name of the descriptive statistic? | 197. What is the name of the inferential statistic? |
| 196. What is the value of the descriptive statistic? | 198. What is the value of the inferential statistic? |
| | 199. What is the p value (exactly as printed)? |

For husbands, what is the standardized association between soft emotion and positive communication after controlling for other variables in the equation?

- | | |
|--|---|
| 200. What is the name of the descriptive statistic? | 202. What is the name of the inferential statistic? |
| 201. What is the value of the descriptive statistic? | 203. What is the value of the inferential statistic? |
| | 204. What is the p value (exactly as printed)? |

For husbands, what percent of the variance in Y was explained when just the first set of variables were entered on step one?

- | | |
|--|---|
| 205. What is the name of the descriptive statistic? | 207. What is the name of the inferential statistic? |
| 206. What is the value of the descriptive statistic? | 208. What is the value of the inferential statistic? |
| | 209. What is the p value (exactly as printed)? |

For husbands, how much additional variance in Y was explained when the second set of variables was added to the equation on step two?

- | | |
|--|---|
| 210. What is the name of the descriptive statistic? | 212. What is the name of the inferential statistic? |
| 211. What is the value of the descriptive statistic? | 213. What is the value of the inferential statistic? |
| | 214. What is the p value (exactly as printed)? |

For husbands, what is the total percent of variance in Y explained by all the predictors taken together?

- | | |
|--|---|
| 215. What is the name of the descriptive statistic? | 217. What is the name of the inferential statistic? |
| 216. What is the value of the descriptive statistic? | 218. What is the value of the inferential statistic? |
| | 219. What is the p value (exactly as printed)? |

Part G: Collinearity

Consider the correlation table depicted below. It includes three potential predictor variables (X1, X2, and X3) and one outcome variable. Assume these correlations are based on a sample of about 100 people. Use the table to answer the questions that follow.

	X1	X2	X3	Y
X1	1.0			
X2	.9	1.0		
X3	.2	.2	1.0	
Y	.4	.4	.4	1.0

If you estimate a regression equation using only X1 and X2 as predictors:

220. Which beta weights (if any) are likely to be significant?
 A. none B. only X1 C. only X2 D. both X1 and X2
221. What would you expect the R to be? (Note: this is asking about R and not R^2)
 A. trivially larger than .4 and not significant
 B. trivially larger than .4 and significant
 C. noticeably larger than .4 but not significant
 D. noticeably larger than .4 and significant

If you estimate a regression equation using only X1 and X3 as predictors:

222. Which beta weights (if any) are likely to be significant?
 A. none B. only X1 C. only X3 D. both X1 and X3
223. What would you expect the R to be? (Note: this is asking about R and not R^2)
 A. trivially larger than .4 and not significant
 B. trivially larger than .4 and significant
 C. noticeably larger than .4 but not significant
 D. noticeably larger than .4 and significant

If you estimate a regression equation using all three predictors (X1, X2, and X3):

224. Which beta weights (if any) are likely to be significant?
 A. none B. only X1 C. only X2 D. only X3 E. all three
225. What would you expect the R to be? (Note: this is asking about R and not R^2)
 A. trivially larger than .4 and not significant
 B. trivially larger than .4 and significant
 C. noticeably larger than .4 but not significant
 D. noticeably larger than .4 and significant

Part H: Written summaries

The output on the next page is from a study investigating variables that predict the amount of exercise that a person gets. In this study, four variables were assessed for each participant.

Target outcome variable:

Exercise is a scale measuring the amount of physical exercise a person did in the last week.

Three predictor variables:

Income is the person's self-reported annual household income.

Addressing Barriers is a scale assessing the extent to which a person has made progress in finding solutions that address potential barriers to exercise.

Goal Clarity is a scale assessing the extent to which a person has specific, reasonable goals for exercise.

Key question: to what extent does a person's income, progress in addressing barriers, and goal clarity predict the level of exercise?

Correlations

		Exercise	income	addressing_barriers	goal clarity
Exercise	Pearson Correlation	1	.078	.235	.385
	Sig. (2-tailed)		.086	<.001	<.001
	N	482	482	482	482
income	Pearson Correlation	.078	1	.098	.142
	Sig. (2-tailed)	.086		.031	.002
	N	482	482	482	482
addressing_barriers	Pearson Correlation	.235	.098	1	.519
	Sig. (2-tailed)	<.001	.031		<.001
	N	482	482	482	482
goal clarity	Pearson Correlation	.385	.142	.519	1
	Sig. (2-tailed)	<.001	.002	<.001	
	N	482	482	482	482

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.388 ^a	.150	.145	5.67

a. Predictors: (Constant), goals, income, addressing_barriers

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2719.63	3	906.54	28.22	.000
	Residual	15357.40	478	32.13		
	Total	18077.03	481			

a. Dependent Variable: Exercise

b. Predictors: (Constant), income, addressing barriers, goal clarity

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	-4.13	.95		-4.36	<.001
	income	.08	.16	.02	.54	.592
	addressing barriers	.06	.07	.05	.95	.340
	goal clarity	.33	.05	.36	7.21	<.001

a. Dependent Variable: Exercise

226. **Write a summary of these regression results** (printed on the previous page). Your written summary should follow the template provided in your notes and described in the lecture add-on video. After you have provided a written summary, use the scoring system in the box below to grade your own answer. On Canvas, you will be asked to report if you wrote this summary and graded your own answer. The scoring system in the box below is the same as the scoring system that will be used to grade written summaries on learning assessments.

Scoring for a written summary of regression results

This question is worth 15 points. Give yourself 1 point for correctly including each of the following components in your summary.

1. The first sentence tells the reader what type of analysis is being reported, and specifically uses the word “regression,” names the outcome variable (exercise), and states that this outcome is being predicted by (or regressed on) income, addressing barriers, and goal clarity.
2. Summary states that the three predictors (must imply plural or say all three) explained a significant percent of variance in wellbeing (this is a statement about the total, omnibus effect. Other wording with the same meaning and similar clarity is OK).
3. Summary reports: $R^2 = .15$
4. Summary reports: $F(3, 478) = 28.22$
5. Summary reports: $p < .001$ (Give half credit if p is listed as equal to .000)
6. Summary refers reader to a table before reporting results from correlations and betas.
7. Summary interprets the correlation between goal clarity and exercise as being both (a) medium, and (b) positive. (Statement must name the two variables and must use the word “correlation.” No credit for a statement that does not clarify what variables are being referred to, or that merely describe this and an “effect.”)
8. Summary interprets the beta for goal clarity by stating that the effect remained significant controlling for other variables. (This statement must be clearly referring to the relationship between social support and wellbeing. You do not need to use the word “beta,” but you must NOT call it a “correlation.”)
9. Summary interprets the correlation between addressing barriers and exercise as being both (a) small, and (b) positive. (Statement must name the two variables and must use the word “correlation.”)
10. Summary interprets the beta for addressing barriers by stating that the effect became NON-significant controlling for other variables. (This statement must be clearly referring to the relationship between addressing barriers and exercise. You do not need to use the word “beta,” but you must NOT call it a “correlation.”)
11. Summary interprets the association between income and exercise by saying these variables were not related to each other (or saying that the correlation and beta for these variables were both non-significant). Must **NOT** say anything in the paragraph that implies that there is any kind of association between these variables. For example, must **NOT** call it a trivial or small effect.
12. The summary is easy to read. Points 1 through 6 are in the order listed above or in some other sensible order. The summary does not contain unnecessary information. It does not contain sentences that are difficult to understand.
13. The summary includes a table with a column for correlations listing values of .08, .24***, and .39*** for income, addressing barriers and goal clarity respectively. It is OK if variables are in a different order. However, the asterisks must be used appropriately.
14. The table also includes a column for standardized betas listing values of .02, .05, and .36*** for income, addressing barriers and goal clarity respectively.
15. The table includes labels for the two columns, names for the variables, and an appropriate note stating the meaning of the asterisks.

Deduct **3 points** if there is a major misinterpretation of the results, such as a mix up about whether results are significant, or a contradictory statement implying that results are both significant and not significant.

Example answer

In regression analysis, exercise was predicted using income, addressing barriers, and goal clarity. The three predictors explained a significant portion of variance in exercise ($R^2 = .11$, $F(3, 478) = 28.22$, $p < .001$). Correlations and standardized betas are reported in Table 1. There was a medium positive correlation between goal clarity and exercise, and goal clarity explained unique variance in exercise after controlling for the effects of the other variables in the equation. Although there was a small positive correlation between addressing barriers and exercise, the effect became non-significant after controlling for other variables. Notably, income was not significantly related to exercise.

Table 1
Predicting exercise

	Correlation	Standardized beta
Goal clarity	.39***	.36***
Addressing barriers	.24***	.05
Income	.08	.02

*** $p < .001$

References

- Sanford, K. (2006). Communication during marital conflict: When couples alter their appraisal, they change their behavior. *Journal of Family Psychology*, 20, 256-265.
- Sanford, K. (2007). Hard and soft emotion during conflict: Investigating married couples and other relationships. *Personal Relationships*, 14, 65-90.